
TITLE PLACEHOLDER: THE PHYSICS & BIOLOGY OF EMERGENT STRUCTURES *

ABSTRACT

Keywords Artificial life · Emergence · Simulations · Representation learning

1 Introduction

The evolution of a world, conditioned on its initial state, can be described by rules at different scales, from microscopic to macroscopic rules. Take our world as an example. At the microscopic scale, physics describes fundamental forces and particles. Then, zooming out, chemistry describes molecular structures and chemical bonds. At an even more macroscopic scale, biology describes cells, tissues, and organisms. Biological systems exhibit emergent structures that arise from the interactions of particles, but such emergence is hard to predict from the movements of particles due to its entirely new behavior, as Anderson stated in his “More is Different” paper [1]. Thus, there are two problems that are interesting to explore:

1. Can we distill emergent structures of a given world?
2. Given the “physics” rules of a world, can we predict its “biology” rules, and vice versa?

Emergent structures are hard to foresee as they are not obvious from the microscopic level, so we might only recognize them once they arise. This is perhaps related to the idea of computational irreducibility, which states that the only way to know the future state of a complex system is to simulate it step by step, without any shortcut available. However, if we only care about some coarse-grained properties, then we might be able to foresee the future state and thus distill the emergent structures.

Such distillation of emergent structures constitutes what we call the “biology” rules of a world, which are the macroscopic rules that govern the emergent structures. Given the microscopic “physics” rules and the macroscopic “biology” rules, can we determine one from the other?

As a testbed, we picked an artificial life (ALife) substrate, Game of Life. Game of Life “physics” rules are easy to learn as they just involve convolution, but to predict the emergent structures (its “biology” rules) is a non-trivial task, as the system is very chaotic and a tiny difference in initial state can result in very different states in a few steps.

2 Related Work

2.1 Emergence

Emergence refers to phenomena where some properties arise from a complex system that are not present when only looking at the individual components. There are many subfields of emergence, of which two ideas are highly relevant to our work.

Causal Emergence The idea of causal emergence [2] claims that macroscopic dynamics have more causal power than microscopic. For example, neurons in a human brain (microscopic) give rise to the emergent structure of the overall

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brain (macroscopic), and in turn, the brain can cause neurons to activate. In other words, the macroscopic structure that arises from emergence can in turn have more causal power on the microscopic structure. This might give us reason to believe that if we have macroscopic rules, we can predict the corresponding microscopic rules.

Computation & Emergence Crutchfield [3] claims that when emergence happens, the minimal program required to describe the system will both become shorter and switch to a new computational model. Take the emergent behavior of an ant colony as an example. The behavior of an individual ant is just random walk which can be described with a finite-state machine, and the program would be very long if we want to track each ant’s state. If we instead consider the ant colony as a whole, ants prefer to follow stronger pheromone trails, and such behavior can be described with a pushdown automaton that would be much shorter, as it just needs to encode the pheromone map. Some work from causal emergence claims that when causal workings are maximized, time reversal symmetry, or reversibility, is maximized, which might be linked to the idea that the system can then be described with a shorter program.

3 Results

3.1 Distill Emergent Structures from Observations

We hypothesize that the capability to distill a world’s emergent structures might correlate with its capability to predict a future state many steps ahead, as emergent structures often operate at a more coarse-grained level than the microscopic particles. Hence, to distill the emergent structures of Game of Life (GoL), we adopted the task of predicting a GoL state many steps ahead. GoL has clear deterministic rules, and in particular, Conway’s GoL is Turing-complete. Predicting the immediate next state ($t \rightarrow t + 1$) is an easy task, but predicting multiple steps ahead ($t \rightarrow t + \Delta t$, where $\Delta t > 1$) is a much harder task. Thus, we expect that a model has to compress information more efficiently at a higher level as Δt increases. To test on a variety of simulations not limited to Conway’s GoL, we also used other GoL rules found by the ASAL framework by [4].

Model We trained ConvNet that maps from GoL state t to state $t + \Delta t$ in one forward pass. Specifically, the ConvNet takes a 2D GoL grid as input, encodes it, and uses a 1×1 convolution to put it into a D -dimensional feature map. It adds a learned embedding for Δt , the number of timesteps skipped ahead, to each spatial location. Then, it applies a series of residual blocks. Finally, a 1×1 convolution projects back to the output channels.

Representation Space Evaluation To measure whether the ConvNet’s representations capture the emergent structures in GoL, we use a linear probe from the representations of input GoL frame to CLIP vector encoding the frame. CLIP encoding could be seen as the average of lots of humans’ representations, given its training data, so quantifying how well the representations linearly map to CLIP vectors is more objective than evaluation metrics such as how well the representations encode the number of gliders.

Results Preliminary results show that models asked to skip more timesteps ahead (higher Δt) have better representations of input states (states t), which is reflected as linear probe having higher R^2 to CLIP vector (as shown in the middle panel of Figure 1). The representations of target states (states $t + \Delta t$) don’t necessarily improve as Δt increases (as shown in the right panel of Figure 1). But when compared against the baseline of linear probing a random-weights model, the representations of target states do surpass the baseline’s as Δt increases.

To analyze the quality of representations, we also visualize models’ predictions. For example, Figure 2 below shows $\Delta t = 16$ model’s performance on GoL 7291. When comparing the model’s prediction in the third row to the ground-truth target states in the second row, we can see that the model is learning to capture abstract patterns more confidently, without caring too much about predicting each cell precisely in the more chaotic regions. As an analogy, the model captures the "organisms" in GoL, without caring too much about what molecules or atoms are present. This is exactly what we hope to see from a model that learns to capture the "emergent structures" of GoL.

3.2 Distill Emergent Structures from “Physics” Rules

In the task above, we distilled emergent structures from observations of GoL trajectories. An interesting follow-up question is whether we can distill emergent structures directly from the “physics” rules of GoL, without ever needing to observe any trajectories. CLIP vectors encoding GoL states could be seen as the emergent structures of GoL, as there has been work showing that CLIP vectors capture high-level information but not low-level details [need to find the paper to cite]. For instance, CLIP vectors can capture e.g. that there will be trees in this part of the image, but not the exact number of leaves on each tree. The concrete task then is, given a vector encoding a GoL world’s “physics” rules, can we predict the CLIP vector encoding the emergent structures of that world?

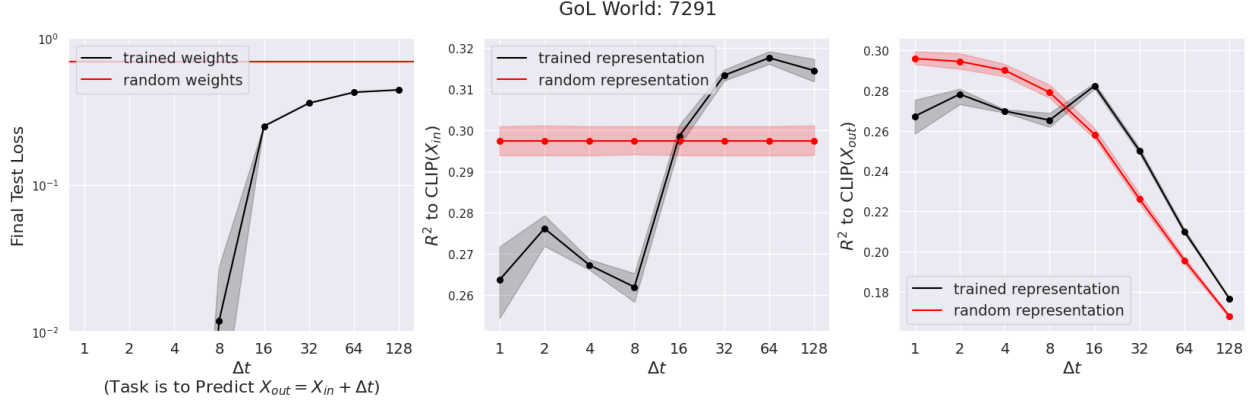


Figure 1: Results on Game of Life 7291. Left: ConvNet training result. Middle: Linear probe result for input state. Right: Linear probe result for target state.

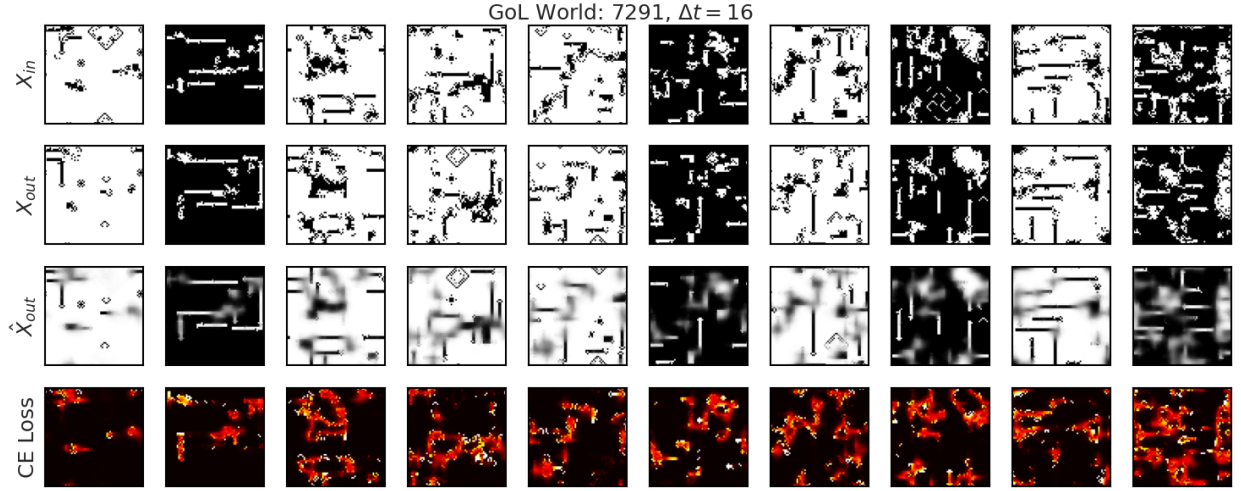


Figure 2: ConvNet performance on Game of Life 7291, with $\Delta t = 16$. First row: Input state. Second row: Ground-truth target state. Third row: Model prediction of target state. Fourth row: Cross-entropy loss.

“Physics” Rules One world maps to one set of “physics” rules. The different worlds toggle on / off different rules, such as “a dead cell with 9 alive cells around it becomes alive”. There are in total 2^{18} possible worlds. In implementation, a set of “physics” rules is a vector whose entries specify whether a cell will be alive or dead, given its current state and its neighbors’ states.

Model We trained an MLP that takes a vector encoding the “physics” rules of a GoL world as input, and outputs a CLIP vector encoding a state from that GoL world.

Results Preliminary results show that it is possible to predict the emergent structures of a world just from its “physics” rules, without even needing to observe any trajectories. Previously, ASAL [4] sorted the 2^{18} possible worlds by their open-endedness score, which basically ranks the worlds by how emergent they are. We trained the MLP on the most emergent worlds. As shown in Figure 3, as the model is trained on more worlds, the model gets higher R^2 on predicting the CLIP vectors for the different worlds.

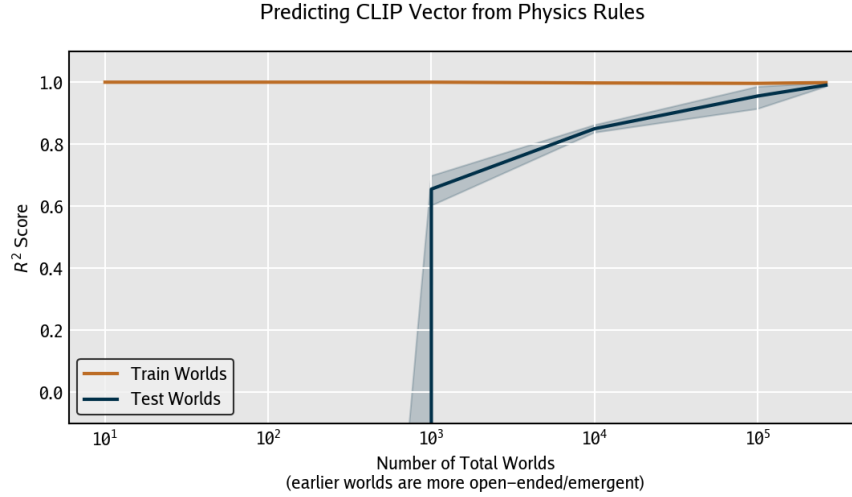


Figure 3: MLP’s R^2 score on predicting CLIP vectors for different GoL worlds, given the worlds’ “physics” rules. As the model is trained on more worlds, the model becomes better at predicting the CLIP vectors, i.e., distilling the emergent structures of the worlds.

3.3 Predict “Biology” Rules from “Physics” Rules

3.4 Predict “Physics” Rules from “Biology” Rules

4 Discussion

References

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